INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

Governance, Compliance, and Risk (GCR)
Phase 2, Course 5: GRC205 - Incident Management and Business Continuity

Business Continuity and Disaster Recovery Assignment: on "Northbridge Skills Institute"

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Learning Outcomes:

Through the analysis and completion of this assignment, the following core capabilities were demonstrated and applied in this report:

1. **Business Impact Analysis & Function Prioritisation:** Conducted a Business Impact Analysis (BIA) to evaluate operational dependencies and systematically identify and prioritise critical business functions.
2. **RTO & RPO Justification:** Justified Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO) based on operational risk tolerance and specific business requirements.
3. **Site & Strategy Evaluation:** Evaluated and compared alternative recovery sites and backup strategies, balancing technical capabilities against financial and resource costs.
4. **Resilience Strategy Coordination:** Coordinated roles, workflows, and governance structures across business continuity, disaster recovery, and incident response domains.
5. **Validation & Continuous Improvement:** Tested key recovery assumptions through scenario testing and translated exercise findings into actionable, measurable operational improvements.

ABSTRACT

In this assignment exercise, I conducted a **Business Impact Analysis (BIA) and evaluated the Business Continuity Planning (BCP)** and **Disaster Recovery (DR)** capabilities of Northbridge Skills Institute following a major site outage caused by flooding and power failure. I began by establishing a five-function BIA (F1–F5) to assess operational dependencies, quantify incremental disruption costs over 8-hour and 24-hour periods, and establish priority recovery sequences based on strict institutional deadlines, such as scheduled examinations and payroll submissions. Next, I defined and justified **Recovery Time Objectives** (RTO), **Recovery Point Objectives (RPO)**, and **Maximum Tolerable Downtime (MTD)** targets, incorporating operational buffers and designing an enhanced backup schedule following the 3-2-1 principle with immutable storage to resolve checkpoint gaps.

I then evaluated Cold, Warm, and Hot recovery strategies against cost constraints, technical capabilities, and risk factors, recommending a Warm site strategy aligned with the **NGN 3,000,000 budget limit**. To support operational readiness, I developed a coordinated **BC/DR Plan featuring clear activation thresholds,** a **single-accountability RACI matrix,** a **72-hour temporary operational procedure** for critical functions, a technical DR runbook, and a multi-stakeholder communication plan. Finally, I modeled a baseline recovery schedule, executed a simulated 45-minute tabletop stress exercise with injects, mapped outputs to **NIST SP 800-34 and ISO 22301 frameworks**, and established measurable Key Performance Indicators (KPIs) alongside a 12-month governance roadmap.

Overall, this assignment deepened my practical knowledge of structuring resilient business continuity frameworks, balancing recovery requirements against financial limits, and executing coordinated disaster recovery protocols in an educational institution setting.

Scenario Introduction and Evidence

Business context and impact data

Northbridge Skills Institute

Northbridge is a fictional education provider with 1,800 learners and 90 staff. At 08:00 WAT on Day 0, a Monday, flooding and a power failure make the main site and its local systems unavailable. Facilities estimates that the site will remain inaccessible for 72 hours. No cyberattack has been confirmed. You coordinate continuity and recovery from the time of disruption.

Microsoft 365 identity and email remain available through mobile internet. Local applications use a separate local directory, AUTH01. An alternate facility can provide 25 workstations from 10:00. Of the 120 learners scheduled for an 11:00 examination, 40 need centre workstations and 80 can use their own devices. Two technicians are available for disaster recovery. Business coordinators can organise temporary services and learner support.

Functions requiring a BIA

Function and local systems	Operational constraint and impact	NGN per hour
F1 Teaching and assessment LMS01 and DB01	The examination starts at 11:00. If no acceptable assessment service is available by then, add a one-off rescheduling cost of NGN 120,000. Learner access and assessment integrity matter.	60,000
F2 Fees and finance FIN01 and DB01	Fee confirmation is due by 16:00 on Day 0. The external payment provider remains available, but local receipts and wallet records cannot be updated. Reconciliation is necessary.	40,000
F3 Admissions and registration REG01 and DB01	Admissions close at 18:00 on Day 0. New applications and registration changes cannot be entered into the local system.	20,000
F4 Staff payroll HR01 and DB01	The bank submission deadline is 12:00 on Day 1. Incorrect or duplicate payroll instructions could affect staff and require manual correction.	10,000
F5 Records and certificates FS01	Certificate and record requests have a 48-hour internal service commitment. Files must retain integrity and access restrictions.	5,000

For Task 1, treat the hourly amounts as separate, non-overlapping incremental disruption costs. Apply them for the first 24 hours and add the examination rescheduling cost once when its trigger is met. Calculate the 8-hour and 24-hour impact assuming no recovery or continuity measures. These are exercise estimates, not observed losses.

Dependencies and minimum service

F1 to F4 require the recovery network, AUTH01, DB01, and their respective applications. F5 requires the network, AUTH01, and FS01. All functions also depend on available staff, secure access, and communication. Define a workable minimum service for each function, including the approval and record controls for any manual process.

Recovery evidence and planning options

Available backup checkpoints

At the 08:00 disruption, the offsite backup catalogue contains the following items. Task 5 uses a baseline drill in which these items pass integrity checks. The tabletop injects later tests what happens when that assumption fails.

Item	Data checkpoint WAT	Recovery dependency
B01 DB01 full backup	23:00, Day minus 1	Base for all four business applications.
B02 DB01 incremental chain	07:00, Day 0	Requires B01 and every required incremental through 07:00. This is the latest recoverable business-data point.
B03 AUTH01 snapshot	00:00, Day 0	Local directory and configuration state.
B04 FS01 full backup	23:00, Day minus 1	Record and certificate files.

A further disconnected full backup exists at 17:00 on Day minus 3. Production and network backup administration share an administrator identity. The offsite provider holds the encrypted copies, but its access-independence controls are undocumented. Investigate these weaknesses in your backup design.

Next year strategy options

The annual planning limit is NGN 3,000,000. Each package covers all five functions and shared infrastructure. Add NGN 400,000 annually for independent protected backups and NGN 200,000 for training and tests. Package timings are estimates requiring validation. Use the current drill timings on the next page for Task 5.

Package	Annual base cost NGN	Estimated first critical service	Data-loss window
Cold	600,000	24 to 48 hours	Up to 24 hours
Warm	1,500,000	3 to 5 hours	1 to 6 hours
Hot	2,400,000	About 1 hour	Up to 15 minutes

Recovery drill data and business impact

Current warm site drill timings

Start at 08:00. Each action occupies one technician for its full duration. Two actions may run in parallel only when all prerequisites are complete and technicians are free. Actions cannot be split. Durations include their local validation; a function is available after its application or file action completes.

Action	Minutes	Prerequisite
Recovery network	30	None
AUTH01 restoration	30	Recovery network
Security and backup checks	30	AUTH01 restoration
DB01 restoration	90	Security and backup checks
LMS01 restoration	45	DB01 restoration
FIN01 restoration	30	DB01 restoration
REG01 restoration	30	DB01 restoration
HR01 restoration	45	DB01 restoration
FS01 restoration	60	AUTH01 restoration and security and backup checks

Initial evidence pack

ID	Time WAT	Source and observation
E01	Day minus 1 16:40	User report and mail record: a finance officer opened an unexpected payroll attachment on FIN-PC07. Execution of malicious code is not yet verified.
E02	Day 0 01:15	Microsoft 365 audit: the finance officer account signed in from an unfamiliar IP and new device. MFA status and the user's involvement are unknown.
E03	Day 0 01:35	Windows logs: svc_backup made remote administrator logons to FS01 and BAK01 outside its usual schedule. The reason is unknown.
E04	Day 0 01:50	Proxy record: FIN-PC07 uploaded 1.2 GB over HTTPS to unfamiliar cloud storage. The record does not reveal the transferred content.
E05	Day 0 02:00	Backup console: the FS01 and LMS01 job reports success. Recoverability and freedom from compromise have not been established.
E06	Day 0 08:30	EDR and helpdesk: mass file changes occur on FIN-PC07. FS01 shared folders contain .locked files and a ransom note. The malware family is unconfirmed.

ID	Time WAT	Source and observation
E07	Day 0 08:40	Helpdesk: the portal login page loads, but learners cannot open materials or submit assessments. The scope of damage to LMS01 remains unknown.
E08	Day 0 08:45	Operations report: the incident contact list on FS01 is inaccessible. Staff phones are available. The trustworthiness of corporate email is unverified.

These observations do not establish attacker identity or prove that personal data was stolen. Cite evidence IDs in your decisions and label additional assumptions. Use fictional contact details and record proposed evidence collection as planned.

My Task:

Act as the Business Continuity and Disaster Recovery Coordinator for a fictional training institute whose main site becomes unavailable. Use the supplied data to assess business impact, set recovery objectives, select an affordable strategy, and demonstrate whether the proposed recovery sequence can meet business needs.

My Response

Assumptions and framework I used

This report is written from the position of the Business Continuity and Disaster Recovery (BC/DR) Coordinator of Northbridge Skills Institute. At 08:00 WAT on Day 0 (a Monday) flooding and a power failure made the main site and its local systems unavailable for an estimated 72 hours. No cyberattack is confirmed. All names, figures and systems are fictional, and every calculation uses only the supplied data. Where the brief does not give a fact, I stated an assumption and marked anything I could not verify as "unverified" so that the reader could see what the plan depends on.

The work follows the ICDFA teaching slides on Business Continuity and Disaster Recovery (ICDFA, 2026.), the NIST contingency planning guide (NIST, 2010) and the ISO 22301 business continuity standard (ISO, 2019).

ID	Assumption	Status
A1	Day 0 is a Monday. All times are WAT. The disruption time T0 is 08:00.	Given
A2	The warm site drill timings in the brief are the baseline. Results from them are modelled, not a live test.	Given
A3	Backups B01 to B04 pass integrity checks in the baseline. The Task 5 tabletop tests the case where this fails.	Given
A4	DB01 incremental backups run hourly, so the chain to the 07:00 point holds eight incrementals.	Unverified (U1)

ID	Assumption	Status
A5	The Microsoft 365 licences include a forms or assignment tool that can run a temporary examination.	Unverified (U6)
A6	The approved examination paper can be obtained outside LMS01 (for example from Microsoft 365 or a printed copy).	Unverified (U5)
A7	The alternate facility has room for 120 learners and invigilators. Only its workstations are limited. An examination sitting lasts up to 90 minutes.	Unverified
A8	Role holders named in this report (for example Academic Director, Finance Manager) are fictional job roles.	Assumption

Part 1: Business Impact Analysis

Task 1: Business Impact Analysis (20 marks)

1. Build a BIA covering F1 to F5. For each function, record the owner role, users affected, critical resources, dependencies, impact over time, minimum acceptable service, and proposed recovery priority. Include people, premises, suppliers, and information needs.
2. Calculate the total estimated disruption cost for 8 hours and 24 hours using the supplied rates and the one-off examination trigger. Show the calculation for each function and the combined total. Explain two limitations of these estimates and distinguish financial impact from learner welfare, reputation, and data integrity.
3. Explain how the 11:00 examination and later deadlines affect priorities. Separate business priority from the technical restoration order required by shared infrastructure. Identify at least three important dependency risks.

Required outputs: five-function BIA, worked impact calculations, and a justified priority sequence.

ANSWERS

1.1 Five-function BIA

Table 1.1a: Owners, users, resources and dependencies

Function and systems	Owner role	Users affected	Critical resources: people, premises, suppliers, information	Dependencies
F1 Teaching and assessment (LMS01, DB01)	Academic Director	1,800 learners, teaching staff, and 120 learners in the 11:00 examination	**People:** invigilators, teaching staff, technicians. **Premises:** alternate facility (25 workstations from 10:00). **Suppliers:** alternate facility provider, Microsoft 365. **Information:** approved examination papers, learner list, accessibility adjustments.	Recovery network, AUTH01, security and backup checks, DB01, LMS01; staff, secure access, communication.
F2 Fees and finance (FIN01, DB01)	Finance Manager	Learners paying fees, finance staff	**People:** two finance staff (for dual control). **Premises:** alternate facility or remote access. **Suppliers:** external payment provider, bank. **Information:** provider statements, fee schedule, receipt log.	Recovery network, AUTH01, security and backup checks, DB01, FIN01.
F3 Admissions and registration (REG01, DB01)	Registrar	Applicants, new learners, registry staff	**People:** registry staff. **Premises:** alternate facility. **Suppliers:** Microsoft 365 (forms and email). **Information:** application data, admission criteria, registration changes.	Recovery network, AUTH01, security and backup checks, DB01, REG01.
F4 Staff payroll (HR01, DB01)	HR Manager	All 90 staff	**People:** payroll officer and an approver. **Premises:** alternate facility. **Suppliers:** bank. **Information:** last approved payroll file, approved change forms.	Recovery network, AUTH01, security and backup checks, DB01, HR01.
F5 Records and certificates (FS01)	Records Officer (reports to the Registrar)	Learners and alumni requesting records, staff	**People:** records staff. **Premises:** alternate facility. **Suppliers:** none beyond IT recovery. **Information:** record files, access permissions, request log.	Recovery network, AUTH01, security and backup checks (drill prerequisite), FS01.

Table 1.1b: Impact over time, minimum service and priority

Function	Impact over time (no measures)	Minimum acceptable service and controls	Business priority	Technical restoration position
F1	3 hours (11:00): examination cannot start for 120 learners. NGN 180,000 plus NGN 120,000 one-off. 8 hours: NGN 480,000. 24 hours: NGN 1,440,000. Learner welfare and assessment integrity at risk.	A secure, invigilated examination at the alternate facility: approved paper, photo identification check, signed attendance, time-stamped submissions kept read-only. **Approval:** Academic Director. **Records:** attendance and submission log, later reconciled to LMS01.	1	After DB01, LMS01 is the first application restored (45 minutes).
F2	8 hours (16:00): fee confirmation missed. 8 hours: NGN 320,000. 24 hours: NGN 960,000.	Fee confirmations from the payment provider's records, a restricted manual receipt log, two-person approval, no card data stored. **Approval:** Finance Manager. Reconcile to FIN01 later.	2	After DB01, FIN01 (30 minutes).
F3	10 hours (18:00): admissions close with applications lost. 8 hours: NGN 160,000. 24 hours: NGN 480,000.	Temporary online form or email intake with acknowledgement, and a controlled register for registration changes. **Approval:** Registrar. Duplicate check, then entry into REG01 later.	3	After DB01, REG01 (30 minutes).
F4	28 hours (12:00 Day 1): payroll deadline. 8 hours: NGN 80,000. 24 hours: NGN 240,000. Risk of wrong or duplicate payments.	Payroll prepared only from the last approved payroll and documented approved changes, with two-person check, comparison with the previous run and a duplicate-batch check. **Approval:** HR Manager and Finance Manager.	4	After DB01, HR01 (45 minutes).
F5	48 hours: internal service commitment missed. 8 hours: NGN 40,000. 24 hours: NGN 120,000.	Requests logged in a shared mailbox, urgent ones verified manually, certificates released only from verified files with access restrictions kept. **Approval:** Registrar.	5	Needs only AUTH01 and the checks. Can run in parallel on the second technician.

1.2 Impact calculations

How the impact was calculated. The five hourly rates are treated as separate costs that start at 08:00 and continue with no recovery or continuity measures. Together they add up to NGN 135,000 per hour. Over 8 hours the five functions cost NGN 1,080,000. The one-off examination rescheduling cost of NGN 120,000 is added once, because without any measures there is no acceptable assessment service at 11:00, so the 8-hour total is NGN 1,200,000. Over 24 hours the functions cost NGN 3,240,000 and the same one-off cost gives NGN 3,360,000. The calculation for each function is shown in Table 1.2.

Table 1.2: Disruption cost with no recovery or continuity measures (NGN)

Function	Rate per hour	8-hour calculation	8-hour cost	24-hour calculation	24-hour cost
F1 Teaching and assessment	60,000	60,000 × 8	480,000	60,000 × 24	1,440,000
F2 Fees and finance	40,000	40,000 × 8	320,000	40,000 × 24	960,000
F3 Admissions	20,000	20,000 × 8	160,000	20,000 × 24	480,000
F4 Payroll	10,000	10,000 × 8	80,000	10,000 × 24	240,000
F5 Records	5,000	5,000 × 8	40,000	5,000 × 24	120,000
Subtotal	**135,000**	135,000 × 8	**1,080,000**	135,000 × 24	**3,240,000**
One-off examination rescheduling (trigger met at 11:00)		Added once	120,000	Added once	120,000
Combined total			**1,200,000**		**3,360,000**

Two limitations.

a. First, the rates are fixed exercise estimates, but real harm is not a straight line. It jumps at deadlines such as the 11:00 examination, the 16:00 fee confirmation and the 18:00 admissions close, and it can grow after the first day as learners lose confidence. A flat rate under-states these steps.
b. Second, the model treats each function separately and stops at 24 hours. It ignores shared dependencies, the cost of the response, the backlog after recovery and the rest of the 72-hour closure. The figures help to rank and compare functions, but they are not a forecast of real loss.

Financial impact is not the whole picture. Learner welfare is affected when 120 learners cannot sit an examination, or when learners who need accessibility adjustments are left without them. Reputation depends on whether learners and staff see Northbridge acting fairly and clearly. Data integrity matters because a wrong restore point, a duplicated payment or an altered record can cause harm that is hard to reverse and that no hourly rate captures. For this reason F1 and F4 deserve more attention than their rates alone suggest.

1.3 Priority sequence and dependency risks

Effect of deadlines on priority. The 11:00 examination is only three hours after the disruption, so **F1 is first** by deadline and by cost. Fee confirmation at 16:00 makes **F2 second**, and the 18:00 admissions close makes **F3 third.** Payroll must reach the bank by 12:00 on Day 1, so **F4 is fourth** but has a firm external deadline. The 48-hour service commitment places **F5 last**. This is the business priority order.

The technical restoration order is different, because **shared infrastructure sets the sequence:** recovery network, AUTH01, security and backup checks, then DB01, and only then the four applications. FS01 does not need DB01, so F5 can be restored early on the second technician even though it has the lowest business priority. Business priority decides which application is restored first after DB01 and where a workaround is needed. It cannot shorten the technical chain.

Table 1.3: Business priority and technical order

Business rank	Function	Deadline	Hours after 08:00	Technical restoration order
1	F1	Examination 11:00 Day 0	3	Network, AUTH01, checks, DB01, then LMS01
2	F2	Fee confirmation 16:00 Day 0	8	Same chain, then FIN01
3	F3	Admissions close 18:00 Day 0	10	Same chain, then REG01
4	F4	Bank submission 12:00 Day 1	28	Same chain, then HR01
5	F5	48-hour service commitment	48	Network, AUTH01, checks, then FS01 (parallel to DB01)

Dependency risks. First, DB01 is a single point of failure for F1 to F4. If the B01 and B02 chain fails, four functions fail together. Second, AUTH01 is separate from Microsoft 365 and its snapshot is eight hours old. If it is stale or untrusted, every function is blocked even though email still works. Third, backup administration shares an identity with production and the offsite provider's access controls are undocumented, so one compromised account could harm both live data and the backups. Fourth, only two technicians are available, so any delay on the DB01 path delays all four applications. Fifth, the alternate facility offers only 25 workstations from 10:00, which limits examination capacity.

All Required outputs for Task 1 Completed: Tables 1.1a to 1.3.

Part 1: Recovery Objectives and Strategy Selection

Task 2: Recovery objectives & backup design (15 marks)

1. Define Recovery Time Objective (RTO), Recovery Point Objective (RPO), and Maximum Tolerable Downtime (MTD). Propose and justify numerical targets for each function. State the service level at which you will declare recovery and leave a justified buffer between technical recovery and the tolerable business disruption limit.
2. Calculate the age of the latest recoverable DB01, AUTH01, and FS01 checkpoints at 08:00. Map the resulting potential data-loss windows to the affected functions and compare them with your proposed RPOs. Use the disruption time as the reference point. Explain what can be reconciled from temporary records and what cannot be assumed recoverable.
3. Design a backup schedule that addresses the RPO gaps. Specify full and incremental or differential backups, retention, offsite placement, and the exact chain needed to restore DB01. Apply the 3-2-1 principle and include an independent offline or immutable copy, restricted access, encryption, and restoration tests.
4. Explain why offsite storage or a live replica alone does not establish recoverability. Define how you will check backup integrity, access independence, and the usability of restored data. Label any unverified assumptions.

Required outputs: objective matrix, checkpoint calculations, and backup and restoration design.

Task 3: Recovery strategy and budget (15 marks)

1. Compare the Cold, Warm, and Hot packages against your RTO and RPO targets, staffing needs, geography, supplier dependencies, and the risk of propagating corruption or malicious changes. Treat the quoted recovery windows as planning estimates.
2. Calculate the complete annual cost of each package, including both required additions. Identify which options fit the NGN 3,000,000 limit. Recommend one option, justify the tradeoffs, and identify any objective it cannot meet. Document residual risks and who must accept them.
3. Compare your recommendation with the estimated 8-hour and 24-hour disruption costs. Explain why a single-outage estimate cannot by itself establish annual return on investment. Identify the evidence needed before approving expenditure or accepting supplier recovery claims.

Required outputs: option matrix, itemised budget, and a justified recommendation.

Task 2. Recovery Objectives & Backup Design

2.1 Objectives

Definitions. The Recovery Time Objective (RTO) is the maximum acceptable time from disruption to restoration of a function. The Recovery Point Objective (RPO) is the maximum acceptable data loss, measured in time before the disruption. Maximum Tolerable Downtime (MTD) is the longest outage the business can accept before serious harm, and it includes the RTO plus validation and reconciliation time (ICDFA, teaching slide pg. 6).

Targets and buffers. I set the RTO from the deadline and the modelled recovery time, then keep a buffer to the MTD for validation and reconciliation. For example, F2 has a 16:00 deadline, which is 8 hours after the disruption. An RTO of 4 hours leaves 4 hours for provider reconciliation and approval. F4 has an MTD of 24 hours so that payroll can be checked and approved before the bank deadline at 12:00 on Day 1. The F1 RTO of 4 hours cannot protect the 11:00 examination, because LMS01 is modelled to return at 11:45. That session needs a continuity workaround, and the RTO covers the platform, not that sitting. I will declare a function recovered only when its application or files are restored, technical checks pass and the business owner accepts it against the minimum service in Table 1.1b. Declaring recovery at restoration alone would give false comfort.

Table 2.1: Objective matrix

Function	Tier	RTO	RPO	MTD	Modelled recovery (Task 5)	Buffer to MTD	Service level at which recovery is declared	Justification
F1 Teaching and assessment	1	4 hours	1 hour	8 hours	3 h 45 min	4 h 15 min	LMS01 restored; login, course content and a test submission pass; Academic Director signs.	Platform is needed for the teaching day (8 hours). The 11:00 examination is protected by a workaround.
F2 Fees and finance	1	4 hours	1 hour	8 hours	3 h 30 min	4 h 30 min	FIN01 restored; balances compared with the provider statement; Finance Manager signs.	Fee confirmation is due at 16:00 (8 hours). The buffer is for reconciliation.
F3 Admissions	2	5 hours	2 hours	10 hours	4 h 15 min	5 h 45 min	REG01 restored; application count matches the	Admissions close at 18:00 (10 hours). A 2-hour loss can be

Function	Tier	RTO	RPO	MTD	Modelled recovery (Task 5)	Buffer to MTD	Service level at which recovery is declared	Justification
							intake log; Registrar signs.	re-entered from intake records.
F4 Payroll	2	6 hours	4 hours	24 hours	4 h 15 min	19 h 45 min	HR01 restored; payroll totals match the last approved run; HR Manager signs.	The bank deadline is 28 hours away. An MTD of 24 hours leaves 4 hours for checks.
F5 Records and certificates	3	24 hours	4 hours	48 hours	2 h 30 min	45 h 30 min	FS01 restored with permissions intact; sample files open; Records Officer signs.	The 48-hour service commitment allows the lowest tier.

Shared service milestones: AUTH01 restored within 1 hour of the disruption (modelled 09:00) and DB01 within 3 hours (modelled 11:00). AUTH01 and DB01 have no separate business RTO because they only matter through F1 to F5.

2.2 Checkpoint ages and data-loss windows

Checkpoint ages. Using 08:00 as the reference point, the latest recoverable DB01 point is 07:00 (B02), so its age is 1 hour. The AUTH01 snapshot (B03) was taken at 00:00, so it is 8 hours old. The FS01 backup (B04) was taken at 23:00 on Day minus 1, so it is 9 hours old. The DB01 window of 1 hour affects F1 to F4, because transactions between 07:00 and 08:00 are not in the backup. The AUTH01 window of 8 hours affects every function, because new accounts, password resets and group changes since midnight may be missing. The FS01 window of 9 hours affects F5 only. Against my RPOs, DB01 meets F1 to F4, but with no margin for F1 and F2. AUTH01 misses its 4-hour RPO by 4 hours and FS01 misses its 4-hour RPO by 5 hours. If the incremental chain fails, DB01 falls back to B01 at 23:00, which is 9 hours old and fails every DB01 RPO.

Table 2.2: Checkpoint calculations at 08:00 on Day 0

Item	Checkpoint	Calculation	Age	Functions affected	Proposed RPO	Result
B02 DB01 incremental chain	07:00 Day 0	08:00 − 07:00	**1 hour**	F1, F2, F3, F4	1 h, 1 h, 2 h, 4 h	Meets all. No margin for F1 and F2.

Item	Checkpoint	Calculation	Age	Functions affected	Proposed RPO	Result
B01 DB01 full (fallback if chain fails)	23:00 Day −1	08:00 − 23:00 (previous day)	**9 hours**	F1, F2, F3, F4	1 h, 1 h, 2 h, 4 h	Fails by 8 h, 8 h, 7 h and 5 h.
B03 AUTH01 snapshot	00:00 Day 0	08:00 − 00:00	**8 hours**	F1 to F5	4 hours	Fails by 4 hours.
B04 FS01 full	23:00 Day −1	08:00 − 23:00 (previous day)	**9 hours**	F5	4 hours	Fails by 5 hours.
Disconnected full	17:00 Day −3	3 days before Day 0 at 17:00 to 08:00 = 24 + 24 + 15	**63 hours**	All	Not applicable	Last resort only.

Reconciliation. Some data can be rebuilt from temporary records: fee payments from the external payment provider, applications and requests received by email or temporary forms, manual attendance and examination records, and approved payroll changes held in email. Other data cannot be assumed to be recoverable: any transaction made between 07:00 and 08:00 and not captured elsewhere, local receipts and wallet entries, directory changes after 00:00, file changes on FS01 after 23:00, and anything created on local systems during the outage. The disconnected backup from 17:00 on Day minus 3 is 63 hours old and is a last resort only.

2.3 Backup and restoration design

Backup design and why it works. The schedule in Table 2.3 closes the gaps: DB01 gets hourly differential backups, FS01 gets four-hourly differentials, and AUTH01 is captured every four hours and after any privileged change. I chose differentials for DB01 rather than a long incremental chain because a restore then needs only two sets, the nightly full and the latest differential, and one damaged differential does not break every later point. The design follows the 3-2-1 rule (ICDFA, slide 8) and adds an independent immutable or offline copy, separate backup identities with multi-factor authentication, encryption, and scheduled restoration tests.

Table 2.3: Backup schedule and controls

Element	DB01	FS01	AUTH01
Full backup	Nightly at 23:00	Nightly at 23:00	Snapshot daily at 00:00

Element	DB01	FS01	AUTH01
Incremental or differential	Differential every hour from 00:00 to 22:00	Differential every 4 hours (03:00, 07:00, 11:00, 15:00, 19:00)	Snapshot every 4 hours (04:00, 08:00, 12:00, 16:00, 20:00) and after any privileged change
Worst-case data age	Under 1 hour	Under 4 hours	Under 4 hours
Retention (to be aligned with the records policy)	Differentials 7 days, nightly fulls 14 days, weekly fulls 8 weeks, monthly fulls 12 months	Same as DB01	Snapshots 7 days, weekly 8 weeks
Offsite placement	Encrypted copy of the nightly full and each differential to the offsite provider within 2 hours	Same	Encrypted copy of each snapshot
Independent copy	Weekly full plus latest differential to an immutable (write-once) store or offline media in a separate location	Same	Same
Exact restore chain	**Current:** B01 (23:00 Day −1) plus every incremental from 00:00 to 07:00 in order, which is eight incrementals if they run hourly (U1). **Proposed:** the nightly full plus the latest hourly differential.	Nightly full plus latest differential	Latest valid snapshot
Encryption	AES-256 at rest, TLS in transit, keys held in a separate vault, with an offline key copy held by two named people. Keys are not stored on AUTH01.	Same	Same
Access	Separate backup administrator identity with multi-factor authentication, no use of production administrator accounts, logon alerts, quarterly access review	Same	Same
Restoration tests	Daily checksum and job report, monthly sample restore, quarterly full restore to an isolated environment with an application test, annual parallel test	Same	Same

3-2-1 mapping. Three copies: production, an on-site backup disk, and the encrypted offsite copy. Two media types: disk and cloud storage. One copy offsite. The independent immutable or offline copy is a fourth copy for protection against ransomware and administrator misuse. It is funded by the NGN 400,000 addition.

Why offsite storage or a live replica alone does not establish recoverability. An offsite copy can be incomplete, corrupt, locked by a lost key or controlled by the same administrator account as production. A live replica copies mistakes, corruption and malicious changes within seconds. Recoverability is proven only when a restore succeeds within the RTO and returns data within the RPO. I will check integrity with checksums and a complete-chain check, check access independence by confirming that a production administrator cannot read or delete backup copies, and check usability by comparing record counts and control totals with the provider and by letting each business owner sample the restored data. Assumptions that are not verified are labelled U1 to U6 in Table 2.4.

Table 2.4: Verification checks and unverified assumptions

Check	Method	Pass condition
Integrity	Checksum of every backup file; catalogue check that the full chain is present; malware scan of restored copies	No missing file, no mismatch, no detection
Access independence	Test that a production administrator account cannot read, change or delete backup copies or the provider portal; written control statement from the offsite provider	Access is denied and the provider's controls are documented
Usability	Restore into an isolated environment; compare record counts and control totals; business owner samples records; run an application test	Counts match, sample accepted, test passes

ID	Unverified assumption	How to verify
U1	DB01 incrementals run hourly	Check the backup catalogue
U2	The offsite provider's access is independent of Northbridge administrators	Written controls and a live test
U3	B01 to B04 pass integrity checks (given for the baseline only)	Checksum and restore test
U4	Warm site drill timings match real restoration	Timed trial restoration
U5	The approved examination paper can be obtained outside LMS01	Check the storage location now
U6	Microsoft 365 licences allow a temporary assessment tool	Check licences and run a dry run

All Required outputs for Task 2 Completed: Outputs: Tables 2.1 to 2.4.

Task 3. Recovery Strategy & Budget

3.1 Option comparison

Comparison. The annual cost of each package includes the NGN 400,000 for independent protected backups and the NGN 200,000 for training and tests. Cold costs 600,000 + 400,000 + 200,000 = NGN 1,200,000. Warm costs 1,500,000 + 400,000 + 200,000 = NGN 2,100,000. Hot costs 2,400,000 + 400,000 + 200,000 = NGN 3,000,000.

All three are within the NGN 3,000,000 limit, but Hot uses all of it and leaves no contingency. Cold fits easily but cannot meet any of my objectives: its first service takes 24 to 48 hours and its data loss can reach 24 hours, against RTOs of 4 to 6 hours and RPOs of 1 to 4 hours for F1 to F4. Hot meets the time and data-loss targets, but a live replica also copies corruption or malicious changes quickly, so independent backups are still needed, and Hot needs skilled staff to operate and test it while Northbridge has two technicians. Warm meets most targets and leaves NGN 900,000 unspent.

Table 3.1: Option matrix (quoted timings are planning estimates)

Criterion	Cold	Warm	Hot
Estimated first critical service	24 to 48 hours	3 to 5 hours	About 1 hour
Data-loss window	Up to 24 hours	1 to 6 hours	Up to 15 minutes
F1 and F2 RTO of 4 hours	No	At the lower estimate only; not at 5 hours	Yes
F3 (5 h), F4 (6 h), F5 (24 h) RTOs	No for F3 and F4; F5 only at 24 hours	Yes	Yes
RPO of 1 hour (F1, F2)	No	Only if hourly backups are validated; not at 6 hours	Yes
Staffing	Needs the most build effort from two technicians	Two technicians can run the modelled schedule, with little spare	Needs continuous monitoring skills; two technicians are stretched
Geography	A site must be prepared	Alternate facility must be outside the flood area	Second location in a different flood area, with a link to the main site
Supplier dependencies	Equipment and connectivity delivery	Alternate facility, offsite backup provider, internet provider	Replication provider and facility
Risk of propagating corruption or malicious change	Low, but data is old	Medium: restores from backups, so the independent copy is essential	High: a live replica copies bad changes;

Criterion	Cold	Warm	Hot
			the independent copy is still needed
Annual cost	NGN 1,200,000	NGN 2,100,000	NGN 3,000,000
Within the NGN 3,000,000 limit	Yes	Yes	Yes, with no headroom

Table 3.2: Itemised annual budget (NGN)

Item	Cold	Warm	Hot
Package base cost	600,000	1,500,000	2,400,000
Independent protected backups	400,000	400,000	400,000
Training and tests	200,000	200,000	200,000
Complete annual cost	**1,200,000**	**2,100,000**	**3,000,000**
Planning limit	3,000,000	3,000,000	3,000,000
Headroom	1,800,000	900,000	0

3.2 Recommendation and residual risks

Recommendation. I recommend Warm. It cannot meet the F1 RTO at the top of its estimate (3 to 5 hours against a 4-hour RTO), it cannot meet the 11:00 examination, and its 1 to 6 hour data-loss range meets my 1-hour RPO only if the hourly backups are validated. The examination is therefore handled by a continuity workaround. The residual risks and who must accept them are in Table 3.3. The Principal gives final acceptance on advice from the Executive Steering Committee.

Table 3.3: Residual risks and acceptance

Risk	Why it remains	Treatment	Accepted by
The 11:00 examination cannot be met by restoration (modelled 11:45)	Technical chain is 225 minutes	Temporary examination procedure (Task 4)	Academic Director
Data loss of up to 1 hour, and up to 6 hours at the upper Warm estimate	Backup frequency and package estimate	Hourly differentials, reconciliation from temporary records	Finance Manager, Registrar, HR Manager
Supplier timings are unproven and only two technicians are available	Estimates, not tested results	Trial restoration, external support agreement	Principal on advice of the Executive

Risk	Why it remains	Treatment	Accepted by
			Steering Committee
Corruption or malicious change reaching the backups	Shared administrator identity and unverified provider controls	Separate identities, immutable copy, incident response triggers	Head of IT and Principal
Alternate facility has fewer than 40 usable workstations	Supplier capacity not guaranteed	Multiple sittings, written capacity guarantee	Facilities Manager and Principal

3.3 Cost and return

Cost and return. The no-recovery estimates are NGN 1,200,000 for 8 hours and NGN 3,360,000 for 24 hours, and Warm costs NGN 2,100,000 a year. These figures cannot establish annual return on investment, because one outage is not a year. A return calculation needs the likelihood of such outages each year, the loss that each package really avoids, the yearly cost of owning the package, and the harm that money does not measure. As an illustration, if the Warm baseline restores everything by 12:15, the modelled disruption is NGN 505,000, compared with NGN 1,080,000 for the five functions over 8 hours with no recovery. That avoids about NGN 575,000, or NGN 695,000 if the workaround also saves the examination. This is less than the annual cost, so the decision rests on outage likelihood, learner welfare and data integrity, and these need evidence.

Table 3.4: Illustrative modelled disruption under the Warm baseline (NGN)

Function	Rate per hour	Modelled recovery time	Calculation	Cost
F1	60,000	3.75 hours	60,000 × 3.75	225,000
F2	40,000	3.5 hours	40,000 × 3.5	140,000
F3	20,000	4.25 hours	20,000 × 4.25	85,000
F4	10,000	4.25 hours	10,000 × 4.25	42,500
F5	5,000	2.5 hours	5,000 × 2.5	12,500
Total				**505,000**

Evidence before approval. Before spending money or accepting supplier claims I would require a written service agreement with recovery times and penalties, a timed trial restoration at the warm site, confirmation that at least 40 workstations are guaranteed, proof that the site is not in the same flood area as the main site, independent assurance reports, written access-independence controls from the offsite provider, and outage likelihood data from the facilities risk register.

Part 3: Coordinated Continuity & Recovery Plan

Task 4:Business Continuity & Disaster Recovery Plan (20 marks)

1. Prepare a coordinated plan with scope, activation thresholds, an authorised decision maker and deputy, escalation, contact roles, alternative communication, and situation reporting. Define how the continuity team, technical recovery team, facilities staff, and incident response team work together.
2. Create a RACI matrix for at least six activities, including activation, temporary assessment delivery, backup validation, restoration, stakeholder communication, and return to the primary site. Give each activity exactly one accountable role and at least one responsible role. A combined A/R role is acceptable.
3. Design temporary procedures for F1 to F5 during the 72-hour site outage. Address the 25-workstation limit and 40 learners needing a centre workstation. Explain approvals, accessibility, assessment integrity, privacy, record keeping, and later reconciliation. Identify the minimum staff, equipment, and supplier support required.
4. Write a DR runbook with ordered actions, prerequisites, owners, completion checks, and contingency actions if a step fails. Include backup selection, validation, application and data checks, business acceptance, and rollback. Distinguish restoring technology from resuming an acceptable business service.
5. Provide a stakeholder communication matrix and a 120 to 180 word initial learner notice. Explain when possible compromise would trigger incident response involvement. Define safe return to the primary site, controlled transfer of new records, and final sign-off after facilities clearance and business validation.

Required outputs: coordinated BC/DR plan, RACI, temporary procedures, DR runbook, communication matrix, and learner notice.

ANSWERS

Task 4. Business Continuity & Disaster Recovery Plan

4.1 Scope, activation and roles

Scope and activation. The plan covers loss of access to the main site and its local systems (F1 to F5 and the shared infrastructure), together with the staff, learners and suppliers who depend on them. It is activated when any of these is met: the site or critical systems are expected to be unavailable for more than 2 hours; a function is expected to miss a deadline within 24 hours; people are at risk; or compromise is suspected. Today's event meets the first two. The authorised decision maker is the Principal as Crisis Commander, and the deputy is the Head of Operations. If neither can be reached within 15 minutes, I may start Level 1 actions only. I inform the Crisis Commander within 15 minutes of detection, the Executive Steering Committee within one hour, and the Crisis Commander informs outside bodies such as the insurer. Contact roles are held in a printed list and on staff phones. Communication uses Microsoft 365 first, then SMS and phone calls. Situation reports are issued every 2 hours for the first 12 hours and every 4 hours after that.

Working together. Following slides 12 and 14 of our teaching slide, one Crisis Management Team leads the response. The continuity team runs temporary services, the technical recovery team restores systems, facilities staff manage safety, power, flooding and the alternate facility, and the incident response team stays on call. They share one decision log, one situation report and a joint call every 2 hours. The Crisis Commander settles any conflict.

Table 4.1: Roles, deputies and contact method

Role (abbreviation)	Holder	Deputy	Primary contact	Alternative contact
Crisis Commander (CC)	Principal	Head of Operations	Mobile call	SMS and Microsoft 365 chat
BC/DR Coordinator (BC)	Me	Registrar	Mobile call	SMS and Microsoft 365 chat
Technical Recovery Lead (DR)	Senior technician	Second technician	Mobile call	SMS
Business Owners (BO)	Academic Director, Finance Manager, Registrar, HR Manager	Named deputy in each unit	Microsoft 365 chat	Mobile call and SMS
Facilities Manager (FA)	Facilities Manager	Operations officer	Mobile call	SMS
Communications Lead (CL)	Communications officer	Marketing officer	Mobile call	SMS and social media account

Role (abbreviation)	Holder	Deputy	Primary contact	Alternative contact
Incident Response Lead (IR)	On-call security lead	Backup responder	Mobile call	SMS

Table 4.2: RACI matrix (R = Responsible, A = Accountable, C = Consulted, I = Informed, A/R = same role does both, dash = not involved)

Activity	CC	BC	DR	BO	FA	CL	IR
1. Plan activation	A/R	R	C	C	C	I	I
2. Situation reporting	I	A/R	R	R	R	C	C
3. Temporary assessment delivery	I	R	C	A/R	R	C	I
4. Backup validation	I	I	A/R	I	-	-	C
5. Restoration (DR runbook)	I	C	A/R	C	C	I	C
6. Business acceptance and resumption	I	C	R	A/R	-	I	C
7. Stakeholder communication	A	C	I	C	I	R	C
8. Incident response involvement decision	A	R	R	I	I	I	R
9. Return to the primary site	A	R	R	R	R	I	C

Every row has exactly one accountable role and at least one responsible role.

4.2 Temporary procedures for the 72-hour outage

Temporary procedures. The 11:00 examination is delivered at the alternate facility. Only 25 workstations are available, but 40 learners need one. The 80 learners with their own devices and 25 centre learners sit at 11:00, and the remaining 15 sit at 13:30 on an equivalent paper. Learners with accessibility needs get workstations first. Integrity is protected by controlled paper versions, photo identification, invigilation, time-stamped submissions and a signed attendance record. Procedures for F2 to F5 are in Table 4.3, each with an approver, an access limit and a reconciliation step.

Table 4.3: Temporary procedures for F1 to F5

Function	Procedure	Approval	Accessibility and privacy	Integrity and records	Reconciliation after restoration
F1 Teaching and assessment	Examination at the alternate facility using the approved paper through a temporary	Academic Director	Learners with accessibility needs get workstations and adjustments first, and their	One approved paper per version, photo identification check, about one invigilator per	Attendance and submissions are loaded into LMS01 and a sample of marks is checked

Function	Procedure	Approval	Accessibility and privacy	Integrity and records	Reconciliation after restoration
	Microsoft 365 assessment tool (U6), with printed copies as back-up. Sitting 1 at 11:00: 80 own-device learners and 25 centre learners. Sitting 2 at 13:30: the other 15 centre learners on an equivalent paper (Version B). Second-sitting learners wait in a supervised area with devices stored.		adjustments are recorded. Only names and learner numbers are used. Access to submissions is limited to invigilators and the Academic Director.	25 learners, signed attendance, time-stamped submissions locked read-only, incident log.	by a second marker.
F2 Fees and finance	The payment provider's portal is the source for fee confirmation. Each receipt is entered in a restricted Microsoft 365 log. Fee confirmation is completed before 16:00.	Finance Manager, with a second person checking each batch	Only finance staff have access. No card data is copied.	Numbered log entries, daily provider statements saved read-only.	After FIN01 returns, compare the provider statements and temporary log with FIN01 and wallet records. Investigate every difference and import corrections with two-person approval.
F3 Admissions and registration	Temporary online form or email intake with an acknowledgement to every applicant. Registration changes are written on a controlled register. Admissions close	Registrar	The form works on a phone, and a phone or email alternative is offered. A privacy notice is on the form and only necessary data is collected.	Form timestamps, duplicate check, register signed by the Registrar.	Enter into REG01 and compare counts with the intake log.

Function	Procedure	Approval	Accessibility and privacy	Integrity and records	Reconciliation after restoration
	at 18:00 as planned. If intake fails, the Registrar asks the Principal for a short extension.				
F4 Staff payroll	Prepare payroll only from the last approved run and signed changes. Compare totals with the previous run and check for a duplicate batch reference. Submit by 12:00 on Day 1 only after HR01 is validated. If HR01 is not validated, submit the last approved payroll with only approved changes.	HR Manager and Finance Manager	Payroll data is limited to two named officers.	Batch log, two-person check, comparison sheet.	Compare payments with HR01 records and correct the next run.
F5 Records and certificates	Requests are logged in a shared mailbox with a 48-hour clock. Urgent requests are verified manually. Certificates are released only from verified files after FS01 returns.	Registrar	Identity is checked before any release. Access restrictions are kept.	Issue register with date, requester and approver.	Compare the issue register with FS01 after restoration.

Table 4.4: Minimum staff, equipment and supplier support

Need	Minimum
Staff	Crisis Commander and deputy; BC/DR Coordinator; 2 technicians; 6 invigilators (assumed ratio of 1 per 25 learners plus one spare); 2 finance staff; 2 registry staff; 2 payroll staff; 1 records officer; 1 communications officer; Facilities Manager.
Equipment	25 alternate facility workstations; network equipment and mobile hotspots; stable power; printers and paper copies of examination papers; access to the offsite backup portal; secure storage for papers and logs.
Supplier support	Alternate facility provider (workstations from 10:00); offsite backup provider; internet and mobile providers; payment provider; bank; Microsoft support; insurer; flood and power repair contractors.

4.3 DR runbook

Runbook and acceptance. Restoring technology is not the same as resuming service. A system is technically restored when it passes the checks in Table 4.5. The service resumes only when the business owner confirms that the data is reconciled, access is controlled and the function can meet its deadline.

Table 4.5: DR runbook (modelled baseline timings)

Step	Action	Prerequisite	Owner	Completion check	Contingency if the step fails
1	Activate the plan and brief the team	Disruption confirmed	Crisis Commander and BC/DR Coordinator	Decision logged and roles confirmed	The deputy activates the plan
2	Recovery network (08:00 to 08:30)	Power and connectivity at the alternate facility	Technical Recovery Lead	Routing tests pass and Microsoft 365 is reachable	Use mobile hotspots for Microsoft 365 services, escalate to the internet provider, run continuity procedures only
3	AUTH01 restoration from B03 (08:30 to 09:00)	Step 2 complete	Technical Recovery Lead	Directory service is up, test logons work for each function, group membership sampled against Microsoft 365	Rebuild AUTH01 and recreate critical accounts from Microsoft 365 records and the approved HR list;

Step	Action	Prerequisite	Owner	Completion check	Contingency if the step fails
					use break-glass administrator accounts
4	Security and backup checks (09:00 to 09:30): select the backup, check that B01 and the B02 chain are complete, verify checksums, scan for malware, review the backup administrator logon log	Step 3 complete	Technical Recovery Lead, Incident Response Lead consulted	All checks pass and results are recorded	If the chain breaks, use the last valid point before the break with the business owners' approval of the data loss. If compromise is suspected, call the incident response team. Use the Day −3 disconnected backup only with the Crisis Commander's approval.
5	DB01 restoration (09:30 to 11:00): restore B01, then each incremental in order	Step 4 passed	Technical Recovery Lead	Database consistency check passes, record counts match control totals, applications connect	Restart from the last good point. If it fails, F1 to F4 stay on temporary procedures.
6	FS01 restoration from B04 (09:30 to 10:30), in parallel with step 5	Steps 3 and 4 complete	Second technician	File counts match, permissions test passes, sample files open	Restore an earlier version after approval. Records stay on the manual procedure.
7	LMS01 (11:00 to 11:45)	Step 5 complete	Technical Recovery Lead	Test learner logs in, content opens, test submission is stored	Continue the temporary examination process
8	FIN01 (11:00 to 11:30)	Step 5 complete	Second technician	Balances compared with the provider statement, missing payments listed	Continue the manual receipt log

Step	Action	Prerequisite	Owner	Completion check	Contingency if the step fails
9	HR01 (11:30 to 12:15)	Step 5 complete	Second technician	Payroll totals match the last approved run	Continue the manual payroll procedure
10	REG01 (11:45 to 12:15)	Step 5 complete	Technical Recovery Lead	Application count matches the intake log	Continue the temporary intake form
11	Business acceptance: owners confirm data is reconciled, access is controlled and deadlines are achievable	Steps 7 to 10 and reconciliation	Business Owners	Signed acceptance for each function	Remain on temporary procedures and fix defects
12	Rollback: take a restore point before importing temporary records	Step 11 started	Technical Recovery Lead	Restore point verified	If an import damages data, roll back, correct and repeat; fall back to temporary procedures

4.4 Communication

Table 4.6: Stakeholder communication matrix

Stakeholder	Trigger	Message owner	Channel	Approver	Frequency
Learners (1,800)	Activation, and every service change	Communications Lead	SMS and email, website banner	Crisis Commander	First notice by 09:15, then at least every 4 hours
The 120 examination learners	Examination decision	Communications Lead and Academic Director	SMS and email	Academic Director	By 09:45, then before each sitting
Staff (90)	Activation	Communications Lead	SMS, phone and Microsoft 365 chat	Crisis Commander	Twice daily

Stakeholder	Trigger	Message owner	Channel	Approver	Frequency
Applicants	Admissions intake change	Registrar	Email and website	Registrar	Before 12:00 and at closure
Executive Steering Committee	Activation, and major decisions	BC/DR Coordinator	Phone and email	Crisis Commander	Every 2 hours, then every 4 hours
Alternate facility provider	Activation and capacity changes	Facilities Manager	Phone	Crisis Commander	As needed, hourly on Day 0
Offsite backup provider	Backup access and integrity checks	Technical Recovery Lead	Phone and portal	BC/DR Coordinator	As needed
Payment provider and bank	Fee reconciliation and payroll	Finance Manager	Phone and portal	Finance Manager	Daily
Insurer	Activation	Crisis Commander	Phone	Crisis Commander	Daily

Initial learner notice

Subject: Examination today and temporary arrangements

Dear learner,

Flooding and a power failure have closed the Northbridge main site. We expect it to stay closed for up to 72 hours. Your safety comes first, so please do not travel to the main site.

Your 11:00 examination today will go ahead at our alternate facility. Please bring photo identification. Learners who need a centre computer will receive their sitting time by SMS by 09:45. If you use your own device, charge it and bring your charger. Learners who need accessibility adjustments will be seated first, so please reply to this message with your needs.

Our online learning systems are temporarily unavailable. Please use your Northbridge email and Microsoft 365 account for updates. We will confirm any change by SMS and email.

Do not share your password with anyone. Northbridge staff will never ask for it.

Northbridge Skills Institute, Operations Team

4.5 Incident response involvement and safe return

Incident response involvement. I will call the incident response team if there is any sign of compromise, such as the unexpected backup administrator logon, a failed integrity check, unexplained account changes, or a ransom note. Restoration of affected systems pauses, evidence is preserved, and continuity work carries on (ICDFA teaching slide 14).

Return to the primary site. Return begins when the Facilities Manager certifies the site safe, with flooding cleared, electrical safety checked and power stable. Records created at the alternate facility are frozen at a cut-off time, reconciled, imported by two-person control into the restored systems, and backed up before and after the import. Final sign-off needs facilities clearance, technical validation and business validation, and the Crisis Commander gives the last approval

> **Required outputs completed: coordinated BC/DR plan, RACI, temporary procedures, DR runbook, communication matrix, and learner notice.**

Part 4: Recovery Validation and Program Governance

Task 5:Recovery Schedule and Exercise (15 marks)

1. Using the current warm site timings, produce a schedule with action, technician, start time, finish time, and prerequisite. Respect the two-technician limit. Determine the earliest possible restoration of assessment service and a feasible completion time for all five functions. Show the dependency path that controls the assessment finish time.
2. Calculate each function's elapsed recovery time from 08:00 and compare it with your RTO and MTD targets. Compare restored checkpoints with your RPO targets. Label these as modelled baseline results, not a live test. Explain which deadlines need a continuity workaround even when restoration is successful.

Design and walk through a 45 minute tabletop exercise using these injects. You may simulate all participant roles individually.

Exercise time	Required inject
Minute 10	The site closure extends from 72 hours to five days.
Minute 20	The alternate facility can provide only 15 working workstations.
Minute 30	A DB01 incremental backup fails its integrity check. An unexpected backup administrator logon also requires investigation.
Minute 40	The payment provider reports successful payments absent from the restored finance records.

3. Record a decision and revised action for every inject, including its owner, rationale, and effect on service or recovery. Keep this stress scenario separate from the baseline schedule. Produce at least five corrective actions with priority, owner, target date, and measurable completion evidence.

Task 6:Governance and Performance Measures (10 marks)

1. Assign plan ownership and approval. Propose a risk-based testing calendar with review triggers, training, version control, and distribution. Map at least four of your outputs to NIST SP 800-34 planning steps and four to the ISO 22301 themes covered in the slides. Explain one relevant flood or climate consideration.
2. Define at least four KPIs with formula or measurement method, data source, target, reporting frequency, and accountable owner. Include recovery performance, restoration-test coverage, corrective-action closure, and training. Assess current maturity using the slide 13 model and propose a realistic 12-month improvement target.

Task 5. Recovery Schedule and Exercise

5.1 Baseline schedule (modelled, not a live test)

Schedule logic. After disruption at 08:00, Technician 1 works the critical path while Technician 2 starts FS01 as soon as AUTH01 and the security and backup checks are done at 09:30. The earliest restoration of the assessment service is 11:45. It is controlled by the path recovery network, AUTH01, security and backup checks, DB01 and LMS01, which takes 30 + 30 + 30 + 90 + 45 = 225 minutes. This is after the 11:00 examination start, so a workaround is still needed. After DB01 finishes at 11:00, the four applications need 45 + 30 + 30 + 45 = 150 technician-minutes. Two technicians cannot do better than 75 minutes each, so all five functions are complete at 12:15, which is 255 minutes after 08:00.

Table 5.1: Baseline recovery schedule (Day 0)

Action	Minutes	Technician	Start	Finish	Prerequisite
Recovery network	30	T1	08:00	08:30	None
AUTH01 restoration	30	T1	08:30	09:00	Recovery network
Security and backup checks	30	T1	09:00	09:30	AUTH01 restoration
DB01 restoration	90	T1	09:30	11:00	Security and backup checks
FS01 restoration	60	T2	09:30	10:30	AUTH01 restoration and security and backup checks
LMS01 restoration	45	T1	11:00	11:45	DB01 restoration
FIN01 restoration	30	T2	11:00	11:30	DB01 restoration
HR01 restoration	45	T2	11:30	12:15	DB01 restoration
REG01 restoration	30	T1	11:45	12:15	DB01 restoration

Controlling dependency path for the assessment service: Recovery network (30) → AUTH01 (30) → Security and backup checks (30) → DB01 (90) → LMS01 (45) = 225 minutes, finishing at 11:45. Total technician work is 390 minutes: T1 works 255 minutes and T2 works 135 minutes.

5.2 Comparison with objectives

Comparison with objectives. These are modelled baseline results, not a live test. Table 5.2 shows that every function finishes inside its RTO and MTD, but F1 has only 15 minutes of margin. DB01 is restored to 07:00 (1 hour old), so F1 to F4 meet their RPOs, with no margin for F1 and F2. AUTH01 (8 hours) and FS01 (9 hours) are older than the 4-hour RPO, so the baseline fails there until the new schedule is in place. Even when restoration succeeds, four deadlines still need a workaround. The 11:00 examination falls before LMS01 returns. Fee records for F2 need reconciliation with the provider. F3 applications made after 07:00 cannot be in the restored system. F4 payroll needs a manual check before submission because the data is one hour old.

Table 5.2: Elapsed recovery time and checkpoint comparison

Function	Finish	Elapsed from 08:00	RTO	MTD	RTO met	Buffer to MTD	Restored checkpoint against RPO	Workaround needed
F1	11:45	3 h 45 min	4 h	8 h	Yes (15 min margin)	4 h 15 min	DB01 at 07:00 is 1 h old against a 1 h RPO: met, no margin	**Yes.** The 11:00 examination starts before LMS01 returns.
F2	11:30	3 h 30 min	4 h	8 h	Yes	4 h 30 min	1 h against 1 h: met, no margin	**Yes.** Reconcile payments and receipts since 07:00. The 16:00 deadline can be met.
F3	12:15	4 h 15 min	5 h	10 h	Yes	5 h 45 min	1 h against 2 h: met	**Yes.** Applications made from 07:00 until restoration need temporary intake. The 18:00 close can be met.
F4	12:15	4 h 15 min	6 h	24 h	Yes	19 h 45 min	1 h against 4 h: met	**Control.** Manual check before bank submission. The Day 1 12:00 deadline can be met.

Function	Finish	Elapsed from 08:00	RTO	MTD	RTO met	Buffer to MTD	Restored checkpoint against RPO	Workaround needed
F5	10:30	2 h 30 min	24 h	48 h	Yes	45 h 30 min	FS01 at 23:00 is 9 h old against 4 h: **fails by 5 h**	Rebuild changes since 23:00 from requests and email. The 48-hour commitment can be met.
AUTH01 (shared)	09:00	1 h	1 h milestone	Not applicable	Yes	Not applicable	8 h against 4 h: **fails by 4 h**	Check recent account changes manually.

5.3 Tabletop exercise (stress scenario, separate from the baseline)

Tabletop exercise. The 45-minute exercise tests four injects at minutes 10, 20, 30 and 40. I played each role in turn: Crisis Commander, BC/DR Coordinator, Technical Recovery Lead, Academic Director, Finance Manager, Facilities Manager, Communications Lead and Incident Response Lead. The scenario is a stress test kept separate from the baseline schedule. Its decisions are in Table 5.3b. The main finding is that one sound decision protects the other three: pausing DB01 for investigation protects data, while the continuity procedures keep the examination and payments moving.

Exercise design	Detail
Objectives	Test decisions on a longer closure, reduced workstations, backup integrity and possible compromise, and finance reconciliation.
Roles	Played individually by me: Crisis Commander, BC/DR Coordinator, Technical Recovery Lead, Academic Director, Finance Manager, Facilities Manager, Communications Lead, Incident Response Lead.
Ground rules	Discussion only. No live system is touched. Exercise minute 0 is simulated 09:00 on Day 0 with the baseline schedule under way. Each inject needs a decision, owner and revised action.
Success criteria	A decision is recorded for every inject. The incident response team is called within 10 minutes of the integrity inject. The examination plan is revised within 10 minutes of the workstation inject.

Table 5.3a: Exercise agenda (45 minutes)

Exercise minute	Simulated time	Activity
0 to 5	09:00	Briefing, rules and current status
5 to 10	09:05	Review baseline progress and priorities
10	09:10	**Inject 1:** closure extends from 72 hours to five days
10 to 20		Discuss and decide
20	09:20	**Inject 2:** alternate facility can provide only 15 working workstations
20 to 30		Discuss and decide
30	09:30	**Inject 3:** DB01 incremental fails its integrity check, and an unexpected backup administrator logon needs investigation
30 to 40		Discuss and decide
40	09:40	**Inject 4:** payment provider reports successful payments absent from the restored finance records
40 to 45	09:40 to 09:45	Decide, then record corrective actions

Table 5.3b: Role responses and decisions (simulated)

Inject	Simulated responses by role	Decision and revised action	Owner	Rationale	Effect on service or recovery
1 (minute 10) Closure extends to five days	**Crisis Commander:** continuity must last five days. **Facilities Manager:** the alternate facility must be extended. **BC/DR Coordinator:** temporary procedures must run for	Extend the alternate facility and temporary procedures to six days. Put technicians on 12-hour shifts and ask for external recovery support. Move to a sustained operations plan.	Crisis Commander (A), Facilities Manager and BC/DR Coordinator	The warm site restores systems, but the closure is far longer than any planning assumption. Tired technicians make errors.	Baseline timings are unchanged. Cost and staffing rise.

Inject	Simulated responses by role	Decision and revised action	Owner	Rationale	Effect on service or recovery
	five days. **Technical Recovery Lead:** two technicians cannot work 120 hours without rest.				
2 (minute 20) Only 15 workstations	**Academic Director:** 40 learners need centre workstations and only 15 exist. **Facilities Manager:** will ask the provider to repair or replace faulty machines by 13:00. **Communications Lead:** will update learners.	Sitting 1 at 11:00: 15 centre learners (accessibility first) and 80 own-device learners. Sitting 2 at 13:30: 15. Sitting 3 at 15:30: 10. Use equivalent papers A, B and C. Merge sittings if repairs succeed.	Academic Director (A/R), Facilities Manager	Keeps the examination alive for all 120 learners and protects accessibility. Three sittings end by about 17:00.	More invigilation. NGN 120,000 rescheduling cost is exposed if any sitting slips to another day.
3 (minute 30) Backup failure and unexpected administrator logon	**Technical Recovery Lead:** pauses step 5 and finds which increment failed. **Incident Response Lead:** investigates the logon source and account. **BC/DR Coordinator:** tells owners F1	Pause DB01 restoration. Call incident response. Disable and reset the backup administrator identity and keep logs. Restore to the last valid point before the failed increment only after incident	Technical Recovery Lead with Incident Response Lead. Crisis Commander approves data loss.	Restoring from a tampered or broken chain could bring in malicious or damaged data.	DB01 and F1 to F4 slip beyond baseline. If the 03:00 increment failed, the last valid point is 02:00, which is 6 hours old at 08:00 and misses every DB01 RPO. Temporary procedures cover the gap.

Inject	Simulated responses by role	Decision and revised action	Owner	Rationale	Effect on service or recovery
	to F4 stay on temporary procedures. **Crisis Commander:** authorises incident response involvement.	response clears it and owners accept the data loss. Continue FS01 from its separately verified backup. If not cleared within 60 minutes, consider the Day −3 backup with the Crisis Commander's approval.			
4 (minute 40) Payments missing from finance records	**Finance Manager:** the provider's record is the authority. **Technical Recovery Lead:** checks whether payments fall in the data-loss gap. **Communications Lead:** tells learners that confirmations will follow. **Incident Response Lead:** checks whether records were altered, linked to inject 3.	Treat the provider statement as the source of truth. Do not confirm fees from FIN01 until reconciled. Keep a reconciliation ledger with two-person approval and confirm fees by 16:00 from provider records. Investigate the cause with recovery and incident response, then import corrections after validation.	Finance Manager	Telling a learner that a payment is missing when it is not would cause harm and complaints.	F2 stays on the manual process after FIN01 returns until reconciliation is complete. Finance workload rises.

5.4 Corrective actions

Corrective actions. Table 5.4 lists seven actions with priority, owner, target date and measurable evidence. E is the date of the exercise.

Table 5.4: Corrective action plan

	Action	Priority	Owner	Target date	Measurable completion evidence
1	Separate the backup administrator identity from production, add multi-factor authentication, and obtain written access-independence controls from the offsite provider	Critical	Head of IT	E + 14 days	Access review report, test showing a production administrator cannot reach backups, provider's written controls
2	Change DB01 to a nightly full plus hourly differential and check the chain monthly	Critical	Technical Recovery Lead	E + 21 days	Catalogue shows a complete chain, restore test completes within the RTO
3	Add an independent immutable or offline copy and a quarterly restoration test	Critical	Technical Recovery Lead	E + 30 days	Restore from the independent copy succeeds, report signed by the Head of IT
4	Obtain a written guarantee of 40 working workstations at the alternate facility and test it	High	Facilities Manager	E + 30 days	Signed contract clause, capacity test log showing 40 successful logons
5	Prepare and rehearse a temporary examination pack (Microsoft 365 tool and papers A to C)	High	Academic Director	E + 21 days	Dry run with 20 learners, every submission recorded
6	Write and rehearse the finance reconciliation procedure using provider statements	High	Finance Manager	E + 21 days	Dry run on 50 transactions with zero unexplained differences
7	Agree joint BC/DR and incident response triggers and a staffing	High	BC/DR Coordinator	E + 30 days	Approved procedure and signed support agreement, next tabletop shows

	Action	Priority	Owner	Target date	Measurable completion evidence
	plan for a five-day closure (shifts and external support)				incident response called within 10 minutes

Task 6: Governance and performance measures (10 marks)

6.1 Ownership, testing and maintenance

Ownership and approval. The BC/DR Program Manager (the Head of Operations) owns the plan and keeps it current. The Executive Steering Committee, chaired by the Principal, approves the plan, the budget and any change to recovery objectives, and it meets quarterly (ICDFA, slide 12). Business coordinators own the function-level procedures. The Technical Recovery Lead owns the runbook and the backup tests.

Testing, review and maintenance. Testing is risk-based and follows the frequencies on slide 11, with the highest-risk items tested most often. Table 6.1 schedules quarterly tabletop exercises, a half-yearly walkthrough, an annual simulation and parallel restoration test, and monthly and quarterly restoration tests for DB01, AUTH01 and FS01. A full interruption test is not planned in the next 12 months, because it is high risk for a two-technician team. The plan is also reviewed at once after any test failure, real incident, change of system, supplier or site, change of staff in a key role, or audit finding, and at least once a year (ICDFA, slide 4). Each version has a number, date, approver and change log. The current version is stored in a controlled repository with printed and offline copies at the alternate facility and with each member of the Crisis Management Team, and older copies are withdrawn. Training covers induction for new role holders, role training for the Crisis Management Team and technicians, annual refreshers, and examination continuity training for invigilators.

Table 6.1: Risk-based testing calendar (first 12 months)

Frequency	Activity	Scope and scenario	Owner
Monthly	Sample restore	DB01, AUTH01 and FS01 file or database sample, with checksum report	Technical Recovery Lead
Quarterly	Tabletop exercise	Q1 flood and site loss (this report), Q2 ransomware and backup compromise, Q3 power and network loss, Q4 supplier failure	BC/DR Program Manager

Frequency	Activity	Scope and scenario	Owner
Quarterly	Full DB01 restoration	Restore into an isolated environment with an application test	Technical Recovery Lead
Half-yearly	Walkthrough	Runbook, temporary procedures and contact list	Business coordinators and Technical Recovery Lead
Annually, before the local flood season	Simulation and parallel test	Warm site restoration of all five functions with business acceptance	BC/DR Program Manager
Every 2 to 3 years	Full interruption test	Not planned in the first 12 months. Review in year 3 (ICDFA, slide 11).	Executive Steering Committee

Framework mapping. The mappings in Tables 6.2 and 6.3 are an educational alignment only. They do not claim certification.

Table 6.2: Mapping to the NIST SP 800-34 planning steps (ICDFA, slide 4)

NIST step	Output in this report
1 Contingency planning policy	Plan ownership, approval, authority and testing and training commitments (Task 4 roles, section 6.1)
2 Business impact analysis	Tables 1.1a, 1.1b, 1.2 and 1.3
3 Preventive controls	Backup controls, separate backup identity, immutable copy and flood-aware equipment placement (Tables 2.3 and 6.1)
4 Contingency strategies	Option matrix, budget and Warm recommendation (Tables 3.1 to 3.3)
5 Contingency plan	Activation, RACI, temporary procedures, runbook and communication matrix (Tables 4.1 to 4.6)
6 Testing, training and exercises	Schedule, tabletop and corrective actions (Tables 5.1 to 5.4), testing calendar (Table 6.1)
7 Plan maintenance	Review triggers, version control and corrective-action closure KPI (section 6.1, Table 6.4)

Table 6.3: Mapping to the ISO 22301 themes (ICDFA, slide 9)

ISO 22301 clause	Output in this report
4 Context	Flood and climate consideration, interested parties such as learners, staff, suppliers and the payment provider
5 Leadership	Executive Steering Committee, Crisis Commander authority and deputy (Tables 4.1 and 4.2)
6 Planning	RTO, RPO and MTD objectives and residual risks (Tables 2.1 and 3.3)
7 Support	Training, communication matrix and resource requirements (Tables 4.4 and 4.6)
8 Operation	BIA, strategy, plan, runbook and temporary procedures (Tasks 1 to 4)
9 Performance	KPIs, exercises and management review (Tables 5.3b and 6.4)
10 Improvement	Corrective action plan (Table 5.4)

Flood and climate consideration. Flooding is the cause of this outage, so the plan treats it as a recurring risk and not a one-off. ISO 22301 Amendment 1:2024 asks organisations to consider whether climate change is a relevant issue for their business continuity system (ISO, 2024). For Northbridge this means checking flood risk each year for the main site, the alternate facility and the offsite storage, keeping them in different flood areas, raising critical equipment above likely flood levels, and activating the plan early when a flood warning is issued.

6.2 Key performance indicators and maturity

Table 6.4: Key performance indicators

KPI	Formula or measurement method	Data source	Target	Reporting frequency	Accountable owner
Recovery performance: RTO achievement	Functions restored within RTO ÷ functions tested × 100	Exercise and restoration logs	100% of Tier 1 and at least 90% overall	After each test and quarterly	Technical Recovery Lead
Recovery performance: RPO achievement	Tests where the data-loss window is at most the RPO ÷ tests run × 100	Backup catalogue and restore logs	100% for DB01	After each test and quarterly	Technical Recovery Lead

KPI	Formula or measurement method	Data source	Target	Reporting frequency	Accountable owner
Restoration-test coverage	Critical systems and backup sets restored successfully in the last 90 days ÷ total critical systems and backup sets × 100 (DB01, AUTH01, FS01 and the four applications)	Restoration test log	100%	Monthly	Head of IT
Corrective-action closure	Actions closed by the target date ÷ actions due in the period × 100	Corrective action tracker	At least 90% on time and 100% of Critical actions within 30 days	Monthly	BC/DR Program Manager
Training completion	Role holders trained in the last 12 months ÷ total role holders × 100	Training register	100% for the Crisis Management Team and technicians, at least 90% for invigilators and unit staff	Quarterly	HR Manager

Maturity. Using the slide 13 model, Northbridge is currently at Level 2, Developing. Recovery steps and warm site timings exist for critical systems, but testing is not regular, backup access is not independent, and restoration has not been proven. A realistic 12-month target is Level 3, Defined. That needs documented processes, a tested calendar, clear governance and consistent recovery, which the actions in this report provide. Level 4 is not realistic in a year, because it needs several periods of measured data and links to enterprise risk management.

Table 6.5: Maturity assessment and 12-month improvement path

Measure	Current position	12-month target
Maturity level (slide 13)	Level 2, Developing	Level 3, Defined
Evidence for current level	Basic recovery steps and drill timings exist. Testing is irregular. Backup administration is not	Documented plan approved by the Steering Committee, calendar

Measure	Current position	12-month target
	independent. Restoration is unproven.	completed, KPIs reported quarterly
Quarter 1	Close Critical corrective actions 1 to 3	Independent identity and immutable copy in place
Quarter 2	Tabletop and first full DB01 restore test	All five functions restore-tested
Quarter 3	Annual simulation and parallel test	RTO and RPO achievement reported against targets
Quarter 4	Management review and maturity reassessment	Level 3 criteria evidenced

References

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https://github.com/Chioma-Onyekelu/GRC-Engineering-Labs/blob/main/GRC%20205/GRC205_BCDR_Assignment_2025_GCR_12696_Chioma_Onyekelu.docx.pdf